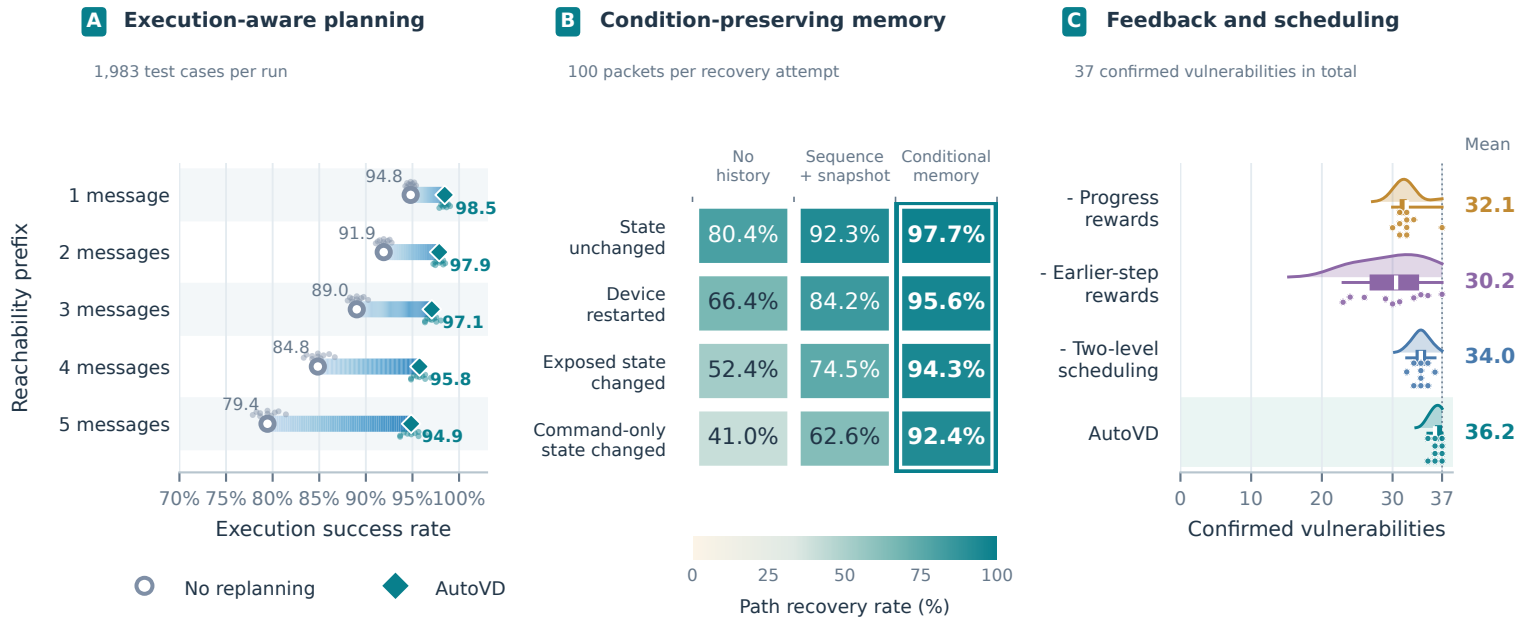

A Execution-aware planning
1,983 test cases per run
Reachability prefix
1 message
2 messages
3 messages
4 messages
5 messages
94.8
98.5
91.9
97.9
89.0
97.1
84.8
95.8
79.4
94.9
70% 75% 80% 85% 90% 95% 100%
Execution success rate
No replanning
AutoVD
B Condition-preserving memory
100 packets per recovery attempt
No history
Sequence + snapshot
Conditional memory
State unchanged
Device restarted
Exposed state changed
Command-only state changed
80.4%
92.3%
97.7%
66.4%
84.2%
95.6%
52.4%
74.5%
94.3%
41.0%
62.6%
92.4%
0 25 50 75 100
Path recovery rate (%)
C Feedback and scheduling
37 confirmed vulnerabilities in total
Mean
- Progress rewards
- Earlier-step rewards
- Two-level scheduling
AutoVD
32.1
30.2
34.0
36.2
0 10 20 30 37
Confirmed vulnerabilities